Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/23/2025
PI Name:	Kolkowitz, Shimon Ph.D.		
Project Title:	Developing New Techniques for Ultra-High-Precision Space-Based Optical Lattice Clock Comparisons		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-7300	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-FP E.6 Fundamental Physics
Start Date:	07/23/2024	**End Date:**	07/22/2027
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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We propose a 3-year, ground-based research program dedicated to developing and demonstrating new techniques that will enable ultra-high-precision clock comparisons between space-based optical lattice clocks (OLCs).
Despite rapid progress in OLC performance over the past two decades, there is a huge performance gap between cutting-edge OLCs operating in research laboratories and space-ready atomic clocks. Lab-based OLCs typically occupy multiple optical tables, consume kW-levels of electrical power, require cooling water, and employ large vacuum chambers with bulky pumps. Portable versions currently under development retain much of the complexity and size, weight, and power (SWaP) of their laboratory counterparts. Simply shrinking down and repackaging existing OLC technologies is unlikely to result in space-based OLCs performing at the levels currently demonstrated in research laboratories. In addition, operation of an OLC in microgravity presents both unique challenges and opportunities. New

Task Description:	and innovative approaches and techniques are required. The PI has recently demonstrated a first-of-its-kind “multiplexed" OLC that enables differential clock comparisons between two or more spatially resolved ensembles of strontium atoms within the same vacuum chamber. These differential measurements eliminate the detrimental effects of clock laser noise and common mode environmental fluctuations, pushing the limits of achievable clock stability and atom-atom coherence. Most recently, we have used this apparatus to perform a blinded, precision test of the gravitational redshift at the millimeter to centimeter scale, and to demonstrate relativistic geodesy measurements with millimeter scale height resolution. Over the course of this 3-year ground-based research program, we will harness the unique capabilities of multiplexed OLCs to explore and demonstrate new techniques to substantially reduce the complexity and SWaP requirements of OLCs, and to enable high performance clock comparisons in space. Our efforts will consist of three main thrusts. The first thrust will focus on ultra-high-precision differential clock comparisons over an optical baseline using "noisy" kHz linewidth lasers, without the need for a frequency comb. The second thrust will focus on demonstrating the use of two ensembles in a single clock to double the operational Ramsey free-precession time and thereby significantly enhance the absolute frequency stability of optical lattice clocks. The third thrust will focus on characterizing and demonstrating a new "magic wavelength" that has the potential to substantially reduce the optical power requirements and the number of lasers required to operate a strontium OLC. Taken together, these three advances will represent a major step towards realizing ultra-high-precision optical clock comparisons in space. The realization of high performance space-based OLC networks would result in major technological impacts on NASA missions and goals. For example, the PI has proposed the use of space-based networks of OLCs linked over optical baselines as gravitational wave detectors with unique capabilities. Such networks could also be used to search for several promising dark matter candidates. Clock comparisons between terrestrial optical clocks and space-based clocks will be sensitive to the height of the Earth-based clocks with respect to the geoid at the centimeter scale, complementing NASA gravitational mapping missions such as the Gravity Recovery and Climate Experiment (GRACE), while comparisons with optical clocks placed in eccentric orbits would enable tests of relativity at entirely new levels of precision. OLCs will enable laser ranging and doppler tracking between satellites at new levels of sensitivity for gravitational mapping and deep space navigation. Terrestrial navigation will also benefit from the enhanced robustness of global navigation satellite systems provided by space-based OLCs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Ultra-high-precision comparisons across space-based clock networks have applications to gravitational wave detection, laser ranging, doppler tracking of spacecraft, gravitational mapping, redefinition of spectroscopic imaging (SI) units, and terrestrial navigation.
Task Progress:	The project team currently consists of Principal Investigator (PI) Kolkowitz, postdoctoral researchers Dr. Shuo Ma and Dr. Dhruva Ganapathy, graduate students Jonathan Dolde and Anke Stoeltzel, and undergraduate researchers Jenny Chen and Bennett Christensen. Over the course of the first year of performance, we have made significant progress towards our goals, and also developed new techniques and research directions that are complementary to our initial proposed approaches. We have already achieved the goals of one of the three proposed primary thrusts of our research program, and have even built upon them. We originally proposed the use of two atomic ensembles in a single atomic clock in order to double the operational Ramsey free precession time without incurring phase slips. We successfully experimentally demonstrated this approach, and confirmed that it does indeed significantly enhance the achievable absolute stability of an optical atomic clock operating with otherwise identical resources, including the same local oscillator and the exact same number of atoms. We demonstrated joint Ramsey interrogation of two spatially-resolved strontium-87 atom ensembles that are 90% out of phase with respect to each other, which we call "Quadrature Ramsey spectroscopy", doubling the achievable coherent interrogation time without incurring phase slips and resulting in a factor of 1.36(5) reduction in absolute clock instability as measured with interleaved self-comparisons. We then went beyond what we had originally proposed, and leveraged the rich hyperfine structure of strontium-87 to realize independent coherent control over multiple ensembles with only global laser addressing. We utilized this independent control over 4 atom ensembles to implement a form of phase estimation, achieving a factor of greater than 3 enhancement in coherent interrogation time and a factor of 2.08(6) reduction in instability over an otherwise identical single ensemble clock with the same local oscillator and the same number of atoms. The manuscript describing this work was published in Physical Review X. [Ed. Note: See Bibliography.] We believe this work represents the start of a new research direction focused on the more efficient allocation and utilization of the resources (both classical and quantum) that make up optical lattice atomic clocks, with the potential for significant improvements in clock stability and accuracy for the same number of atoms, local oscillator, and basic experimental apparatus. We note that we began work on this research thrust immediately after the selection of this grant for funding had been officially announced in March of 2023, but due to administrative delays in the awarding of the grant, we completed this work and published the results before the official start date of the award in July of 2024. However, this work was still enabled by this award and the knowledge that we would be receiving the financial support. The second primary thrust of this research program focuses on determining what ultimately limits the achievable coherence times and instabilities of synchronous differential clock comparisons. Over the first year of performance, we therefore also undertook a detailed and comprehensive set of measurements to develop a complete model of depolarization and atom loss in strontium-87 optical lattice clocks. We performed lifetime measurements for the 1S0 and 3P0 clock states, where we monitored the populations in both states as a function of time, trap depth, and density, in both strontium-88 and strontium-87. We definitively observed radiative decay on the clock transition of strontium-87, and measured the value for the natural radiative lifetime of this transition to be 167 (+79) (-40) seconds. The manuscript describing these results has been submitted to a physics journal and posted to the physics arXiv. The final thrust of our research program focuses on characterizing and demonstrating a new "magic wavelength" "for strontium at 497 nm that has the potential to substantially reduce the optical power requirements for strontium optical lattice clocks. During the first year of performance, we therefore designed, purchased, received, and installed a custom Vertical-External-Cavity Surface-Emitting Laser (VECSEL) laser at 497 nm for this purpose. We expect to begin taking measurements of the 497 nm magic wavelength using this new laser in the coming weeks.

Bibliography Type:	Description: (Last Updated: 07/07/2026)
Articles in Other Journals or Periodicals	Ma S, Dolde J, Zheng X, Ganapathy D, Shtov A, Chen J, Stoeltzel A, Kolkowitz S. "Enhancing optical lattice clock coherence times with erasure conversion." Physics arXiv preprint server. Posted May 9, 2025. https://doi.org/10.48550/arXiv.2505.06437 , May-2025
Articles in Other Journals or Periodicals	Dolde J, Ganapathy D, Zheng X, Ma S, Beloy K, Kolkowitz S. "Direct measurement of the 3P0 clock state natural lifetime in 87Sr." Physics arXiv preprint server. Posted May 9, 2025. https://doi.org/10.48550/arXiv.2505.06440 , May-2025
Articles in Peer-reviewed Journals	Niroula P, Dolde J, Zheng X, Bringewatt J, Ehrenberg A, Cox KC, Thompson J, Gullans MJ, Kolkowitz S, Gorshkov AV. "Quantum sensing with erasure qubits." Phys. Rev. Lett. 2024 Aug 23;133(8):080801. https://doi.org/10.1103/PhysRevLett.133.080801 , Aug-2024
Articles in Peer-reviewed Journals	Zheng X, Dolde J, Kolkowitz S. "Reducing the instability of an optical lattice clock using multiple atomic ensembles." Physical Review X. 2024 Jan 1;14(1):011006. https://doi.org/10.1103/PhysRevX.14.011006 , Jan-2024
Awards	Kolkowitz S. "Experimental Physics Investigator award, The Gordon and Betty Moore Foundation, August 21, 2024." Aug-2024